

Road Accident Analysis

Course: DS/CMPSC 410

Semester: Fall 2025

Team Members: Katherine Thornber, Zachary Smith, Mitchell Darling, Sree Penumuchu, Anant Mishra, Vuk Radulovic

Team Name: I-95

1. Introduction and Motivation

- Provide background on the application domain (e.g., social media analytics, climate data, genomics, finance, IoT).
 - This domain centers on analyzing large datasets of motor-vehicle crashes to uncover how environmental conditions, human behavior, and infrastructure characteristics contribute to accidents. Traffic accident analytics is used by city planners, transportation agencies, and public safety teams to design safer roads, allocate emergency resources, and reduce fatalities. The increasing availability of high-volume traffic and weather data has made this area strongly aligned with big data analytics.
- Identify the key problem/question your project aims to address.
 - From this project we aim to identify the key factors behind severe, accident-prone environments by analyzing: city densities, highway and speed factors and weather impact. The main goal of this project is to identify what causes an severe accident-prone environment. This is essential as it informs safer urban planning, traffic policies, and emergency response.
- Explain why this problem benefits from **big data approaches** rather than conventional data analysis methods.
 - Big data approaches are essential due to the massive volume and high dimensionality of datasets spanning millions of accident records with dozens of features each. This overwhelms traditional analytical tools, especially during complex operations like spatial joins and large-scale feature engineering. These methods are further needed to unravel the complex, multi-factor relationships influencing accident severity, where variables such as weather, traffic flow, time, and road type interact in nonlinear ways that only become visible at scale. Additionally, big data pipelines enable efficient integration and cleaning of heterogeneous multi-source data, merging geospatial, temporal, environmental, and infrastructure information from disparate reporting systems. They also provide the foundation for real-time or

near-real-time processing capabilities, supporting streaming analysis for traffic monitoring and predictive alerts through ecosystems like Spark. Finally, scalable machine learning workflows are critical for training models on millions of records where one can leverage distributed computing to manage intensive tasks such as feature importance analysis, hyperparameter tuning, and model validation, which exceed the limits of conventional single-machine analysis.

2. Project Objectives

Clearly state the goals of the project.

- **Data Acquisition and Initial Processing:** Acquire a large-scale, high-dimensional accident dataset from Kaggle, load it using PySpark, and create a representative sample for efficient initial development and testing in local mode.
 - **Robust Data Preprocessing Pipeline:** Develop scalable PySpark workflows to clean the dataset by handling missing values, standardizing categorical variables, and engineering spatio-temporal features (e.g., time-of-day, season) to prepare data for analysis and modeling.
 - **Spatial-Temporal Analysis via Heatmap Visualization:** Generate geospatial heatmaps to visually identify and communicate areas and time periods with high accident density, pinpointing geographic hotspots and temporal trends.
 - **Network Analysis via Sankey Visualization:** Create Sankey diagrams (sankey plots) to map and visualize the flow of accidents across road segments. This method will approximate vehicle paths to highlight specific highways and roads with the highest accident frequency, overlaying data on state maps for context.
 - **Predictive Modeling and Factor Analysis:** Implement a Random Forest model to predict accident severity. Leverage the model's feature importance scores to quantitatively identify and rank the key contributing factors (e.g., wind speed, presence of traffic signals, temperature) to severe accidents, guiding targeted interventions.
 - **Scalability and Performance Benchmarking:** Systematically evaluate the performance and scalability of the PySpark pipelines and analytical methods by running the workflows with varying numbers of cores on the ICDS cluster, documenting computational efficiency.
-

3. Data Description

- **Data Source:** Sourced from Kaggle: "US Accidents" by Sobhan Moosavi A comprehensive, country-wide collection of traffic accidents from the years 2016-2023. <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>
 - **Data Size:**
 - ~7.7 Million Records (7,728,394 instances)
 - 46 Features per record
 - Total Size: ~3.06 GB
 - **Data Characteristics:**
 - The data is structured (in CSV format), containing:
 - Temporal Features: Date, time, timezone.
 - Geographic Features: Latitude, longitude, street name, city, state.
 - Environmental Conditions: Weather, visibility, sunrise/sunset.
 - Impact Details: Severity, description, distance of traffic delay.
 - **Data Challenges:** Working with an extremely large dataset with high dimensionality, accounting for outliers and missing data, standardizing strings for certain variables.
-

4. Technical Approach

- **Tools & Frameworks:**
 - This project employs a hybrid big-data architecture designed to handle the scale and analytical complexity of the dataset. PySpark serves as the core distributed computing engine for processing and computation. We specifically leverage:
 - Resilient Distributed Datasets (RDDs): For parallelizing data transformations and custom computations that operate efficiently across the cluster.
 - External Python Libraries for Prototyping & Visualization: Pandas, Matplotlib, and Scikit-learn are used for developing and evaluating algorithms locally before scaling them in PySpark.
 - Advanced Geospatial Libraries: GeoPandas, Shapely, Geoplot, and Cartopy for creating sophisticated visualizations, including heatmaps and Sankey diagrams. These tools allow us to process geospatial data samples and render final plots over state maps.
- **Cluster Usage Plan:**
 - Number of Nodes: The analysis was executed on a 4-node cluster, which was necessary to manage the computational demands of RDD operations, geospatial calculations, and model training across the full, high-dimensional dataset.
 - Job Type: All large-scale operations were conducted using batch processing, which was selected to efficiently and reliably process the complete historical dataset.

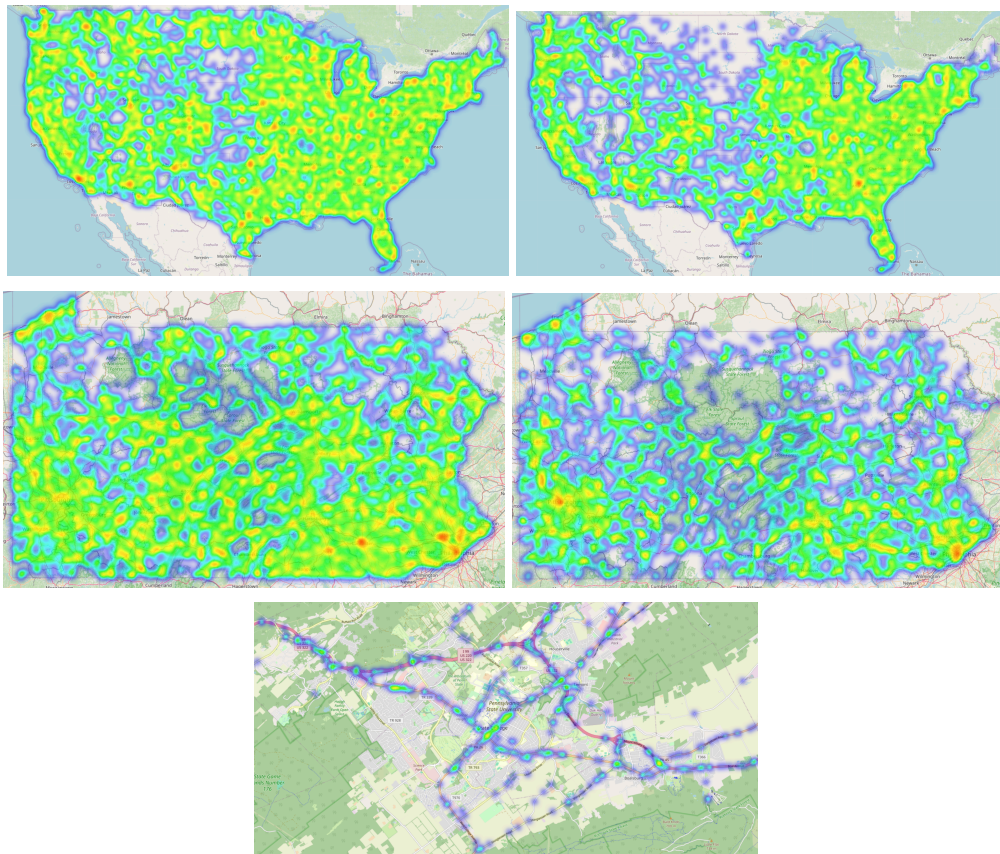
- Storage & File Formats: The following formats were used to store project assets:
 - CSV: For the primary accident dataset.
 - Shapefiles (.shp): To provide geospatial context (state and county boundaries).
 - Notebooks (.ipynb) & Reports (.html): To document the complete analysis pipeline.
 - Images (.png): To save the final visual outputs.
 - **Pipeline Overview:**
 - The analytical pipeline follows a sequential, four-stage workflow, transitioning from data preparation to model insights:
 - Data Ingestion and Preprocessing: The raw dataset is loaded into the PySpark environment. A preprocessing pipeline using RDD transformations is applied to clean the data. This includes imputing or removing missing values and standardizing categorical variables (e.g., grouping similar weather conditions). Key features, such as time-of-day and season, are also engineered at this stage.
 - Exploratory Analysis: In this stage, the preprocessed data is analyzed to uncover initial patterns. This includes calculating summary statistics and performing geospatial aggregation to compute accident densities for specific areas, which provides the foundation for the subsequent visualizations.
 - Model Building and Large-Scale Computation: This phase consists of two primary analytical tracks executed in parallel:
 - Predictive Modeling: A Random Forest model is implemented to predict accident severity. This model is trained and evaluated in a distributed environment to handle the dataset's scale.
 - Visualization Preparation: Aggregated geospatial data is processed into the necessary formats to generate the final heatmaps (showing accident density) and Sankey diagrams (illustrating accident flow along road networks).
 - Visualization and Interpretation: The final step synthesizes and interprets the results. The geospatial heatmaps are rendered to identify accident hotspots. The Sankey diagrams are generated to visualize high-frequency accident corridors. Finally, the Random Forest model's feature importance scores are extracted and analyzed to quantitatively rank the factors most strongly associated with severe accidents.
-

5. Results and discussion

1. What did your data visualization reveal?

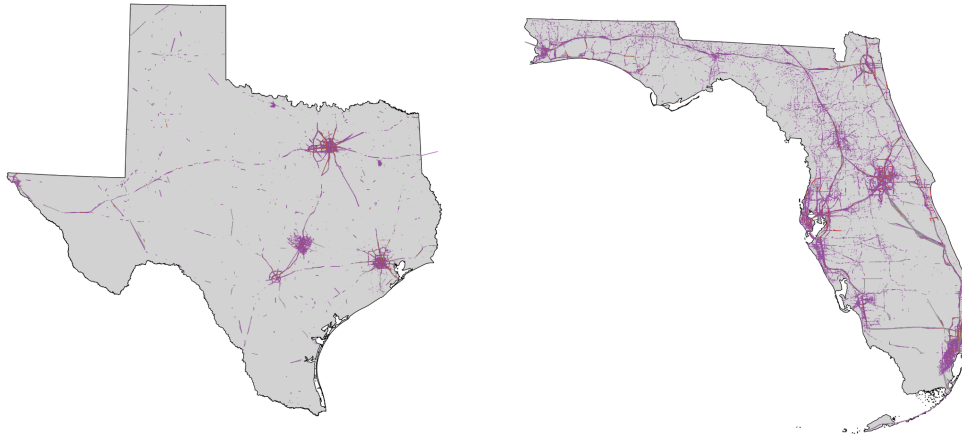
National & State Heatmaps: The heatmap of the continental United States clearly shows that accidents are not uniformly distributed. Major metropolitan areas and their connecting

interstate corridors form pronounced hotspots of high accident density. This visual confirms that population centers and major transportation networks are primary loci of traffic incidents. Zooming into Pennsylvania, the heatmap highlights dense clusters in the Philadelphia and Pittsburgh metropolitan areas. The interstate highways (e.g., I-76, I-95, I-80) connecting these cities are vividly marked as high-frequency accident zones. A focused heatmap of the State College area provides granular insight. It reveals an exceptionally dense concentration of accidents directly on and around the Pennsylvania State University campus and its immediate access roads. A secondary, smaller hotspot is visible at the State Game Lands area, indicating another localized zone of risk. This demonstrates how heatmaps can identify risk at multiple geographic scales, from national corridors to specific local infrastructure.



Images in order (left-to-right): USA heatmap of accidents of all severities. USA heatmap of level 4 severity accidents. PA heatmap of accidents of all severities. PA heatmap of level 4 severity accidents. State College focused heatmap.

Sankey Visualization: The Sankey diagram powerfully illustrates that accidents predominantly trace the path of the U.S. Interstate Highway system within the state. Each line represents an accident origin-destination pair of clusters along these high-speed corridors, particularly between and within major cities. These visualisations conclusively reveal that high-speed highways are the dominant location for accidents, regardless of other contributing factors. It moves beyond point-based heatmaps to show the flow and linear concentration of risk along specific transportation arteries.



Images in order (left-to-right): Sankey diagram of the state of Texas where lines tracing accident paths reveal dense clustering in cities. Sankey diagram of the state of Florida which projects accident flows onto a geographic outline, clearly illustrating that high-speed highways are prominent locations for incidents.

Random Forest Feature Importance: The predictive modeling provided a quantitative ranking of factors influencing accident severity, moving from correlation to attributed importance. Wind Speed (mph) emerged as the most significant factor, with an importance score of 0.316. This indicates that high winds are a major contributor to more severe accident outcomes. Major Contributors: Traffic_Signal (0.196) and Temperature(F) (0.145) were the next most important features. This highlights the critical role of traffic control infrastructure and weather conditions in determining severity. Factors such as Hour (time of day), Visibility(mi), and Precipitation(in) showed measurable but smaller effects. Variables like Rush Hour flags, Low Visibility flags, and FreezingTemp had relatively low importance scores in this model, suggesting that while present, their direct impact on severity is less pronounced than the top factors.

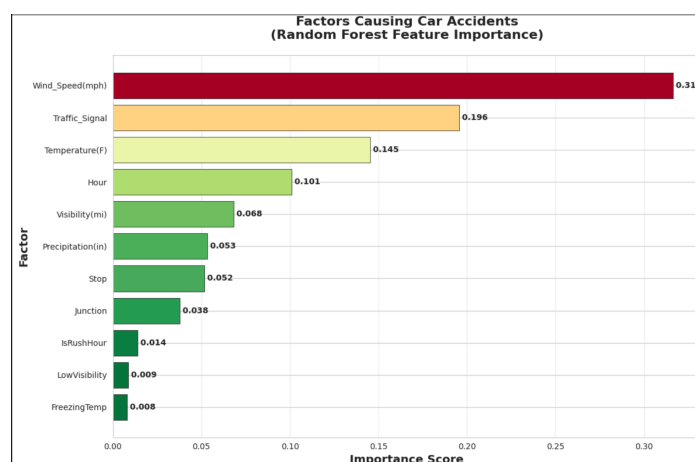


Image above: A horizontal bar chart of Random Forest feature importance scores, identifying wind speed as the most significant factor influencing accident severity, followed by traffic signals and temperature.

2. What did you learn about your data using your model? How did the model perform?

Our analysis revealed clear, actionable insights into the where, how, and why of traffic accidents. The spatial patterns provided by the heatmaps and the Sankey diagrams show that accidents are concentrated on high-speed infrastructure. At a national and state level, heatmaps showed dense hotspots in major cities and along interstate corridors. The Sankey diagram displayed accident paths directly overlaying the U.S. Interstate highway system which proved that these hotspots are continuous and not just isolated spots. These visualizations also proved that the risk of an accident exists at multiple scales from inter-city highways down to specific areas on campus at Penn State. Overall, these visualizations performed exceptionally well. They transformed massive coordinate data into an intuitive, multi-scale narrative of geographic risk. The heatmaps effectively used color gradients to reveal density hotspots at national, state, and local levels. The Sankey diagrams were particularly powerful, using line flow to visually prove that highways are the primary arteries of accident occurrence. Its direct resemblance to a highway map is a strong sign of its validity and clarity. The underlying geospatial aggregations in PySpark demonstrated the ability to handle large-scale data processing, enabling the creation of these insights from millions of records.

The Random Forest model proves that environmental and infrastructural factors dominate severity. The model provided a quantitative ranking, identifying wind speed as the strongest predictor of severe accidents, followed by the presence of a traffic signal and temperature. This indicates a critical interaction between weather and roadway control. Some assumed risks were less influential. Factors like precipitation, rush hour flags, and freezing temperatures had relatively low importance scores in predicting severity, suggesting their role may be more tied to accident frequency rather than outcome. The model succeeded brilliantly as an explanatory tool. The feature importance rankings are logically sound, directly actionable for safety planning (e.g., hardening infrastructure in windy corridors), and fulfilled the goal of identifying key drivers.

The project's multi-method approach was highly effective. The visualizations answered "where and how" accidents cluster, providing the spatial context. The Random Forest model then explained "why" some of those accidents become more severe, identifying the primary contributing factors. Together, they provide a comprehensive, evidence-based foundation for targeted traffic safety interventions.

3. How did your workflow scale on multiple nodes/cpus of Roar?

Our workflow demonstrated clear scalability but also revealed the significant computational demands of advanced geospatial processing. The heatmaps took on a moderate load as the jobs ran efficiently on 2 nodes with 64 GB of memory. The execution took several minutes to complete which indicated that the geospatial aggregation for density calculations was well-distributed and scaled effectively. The task was computationally intensive but manageable with parallel processing. The Sankey visualizations took on a high load and was the most demanding task as it required 4 nodes with 64 GB of memory to process. The execution was very intensive and ultimately led to the job being broken down into smaller batches to be completed. This reveals the heavy computational cost of generating origin-destination flow data for millions of accidents. Calculating pairwise relationships,

creating line geometries, and aggregating them for visualization is a complex, data-shuffling operation that led to the resources given being pushed to the limits.

4. What is the path to your models/code/data on Github?

The code and models are in this Github repository:

<https://github.com/camthorner/ds410carcrash>

The data is too big to upload to the repository, however it can be accessed by this Kaggle link: <https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents>

5. What unsolved problems remain? How would you revisit this course project to address that?

The primary unsolved problem that remains is to achieve predictive, real world impact by shifting from a batch analysis of static data to a more dynamic system that is capable of processing live data streams to generate timely alerts or forecasts. To solve this problem, we would be revisiting the course project with a focus on building a real-time data ingestion and predictive pipeline that takes in live accident, weather, and traffic data, processes it with structured streaming, and feeds it into the deployed random forest model for instant severity prediction. This pipeline can then be connected to an interactive dashboard to display live risk maps and issue automated alerts.

6. CREDIT Statement

Assign roles among team members, using the Elsevier guidance here:

<https://www.elsevier.com/researcher/author/policies-and-guidelines/credit-author-statement>

Mitchell Darling: Conceptualization, Software, Investigation, Data Curation, Visualization, Formal Analysis

Sree Penumuchu: Conceptualization, Validation, Formal Analysis, Writing - Original Draft/Review & Editing, Visualization, Supervision.

Katherine Thornber: Conceptualization, Methodology, Software, Visualization, Formal Analysis

Anant Mishra: Conceptualization, Resources, Investigation

Vuk Radulovic: Conceptualization, Investigation, Resources

Zachary Smith: Conceptualization, Software, Methodology.

8. References

List any datasets, papers, or tutorials you are drawing inspiration from.

Kaggle dataset, US Census Bureau [cartographic boundaries](#), geoplot [examples](#), geopandas [information](#)
